



GUI-Based Multi-Client Chat Application Using TCP

Internship Report

Submitted by

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1. Objective

The objective of this assignment was to transform the terminal-based TCP chat client developed in Assignment 5 into a GUI-based desktop application using Tkinter while reusing the existing TCP server. The implementation provides login validation, broadcast and private messaging, online user management, background message handling using multithreading, and Wireshark verification. The networking logic remains separated from the GUI as required.

2. GUI Design

The application contains a login screen, chat window, online users panel, message area, recipient selector, status bar, Send and Disconnect buttons.

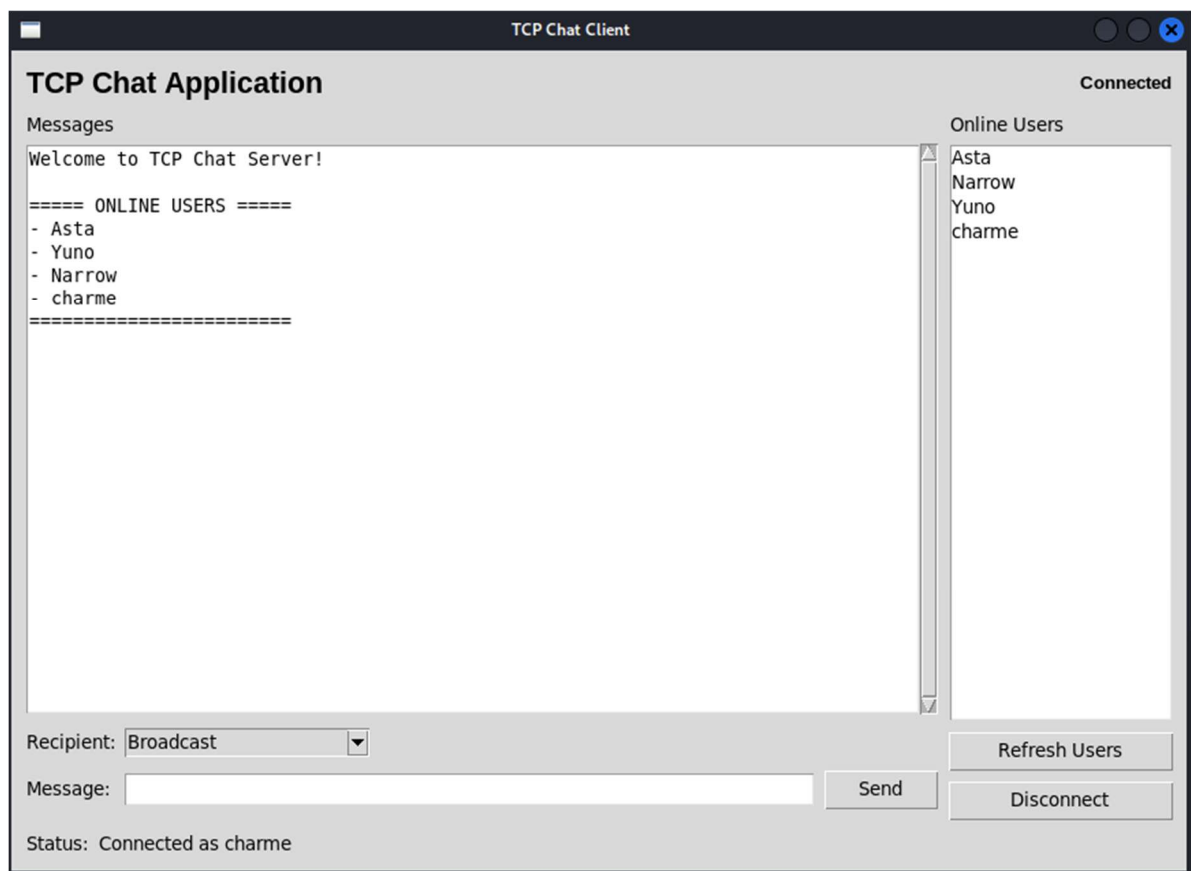


Figure 1: Login and successful connection.

Main Chat Interface

The main interface displays received messages, online users, recipient selection and message input.

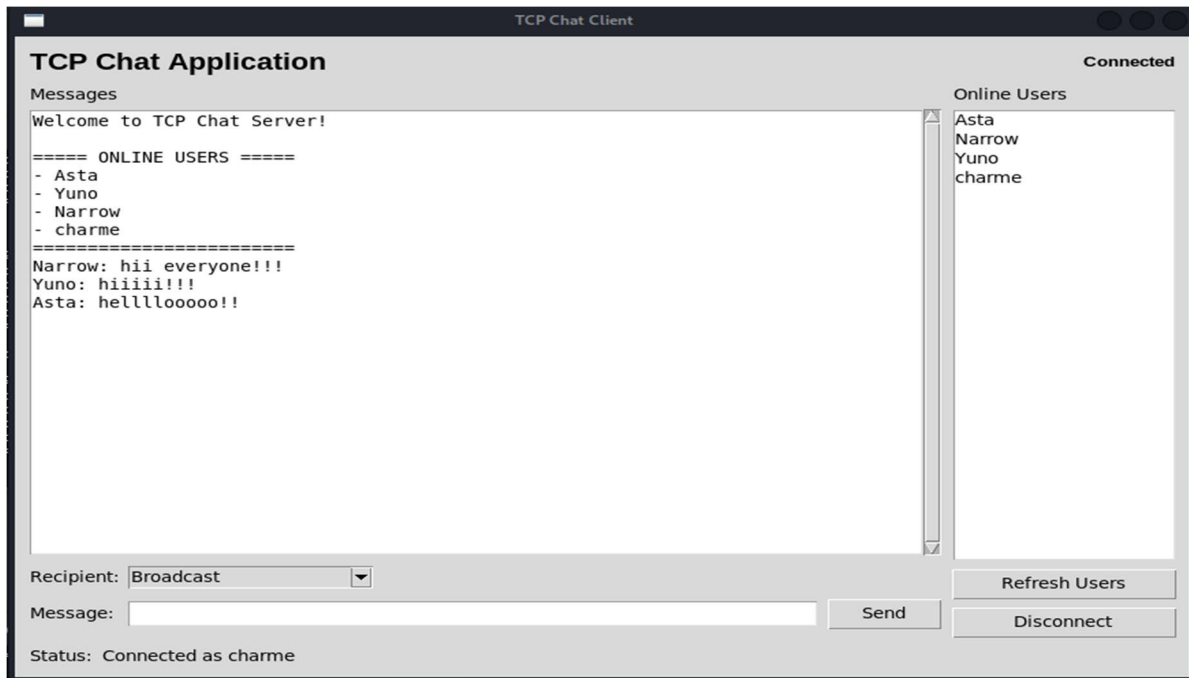


Figure 2: Broadcast messaging window.

Online User Management

The online user list updates automatically whenever users join or leave.

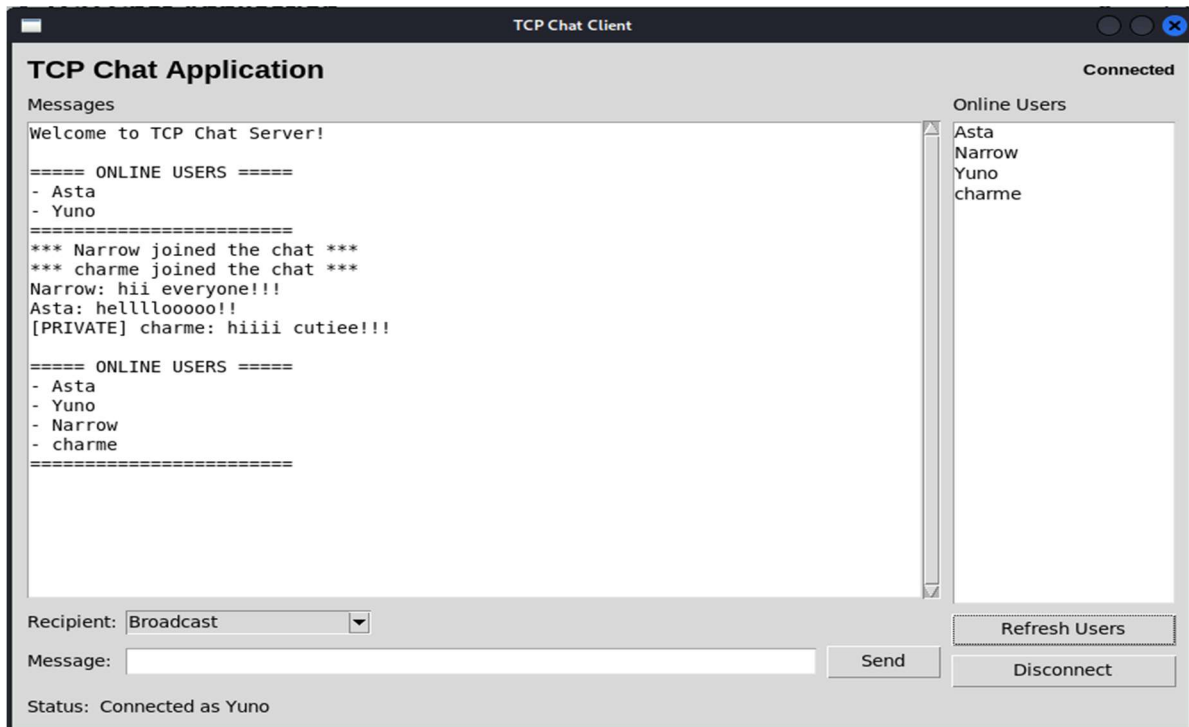


Figure 3: Online users panel.

Private Messaging

Users can choose a recipient and send private messages using the existing server routing.

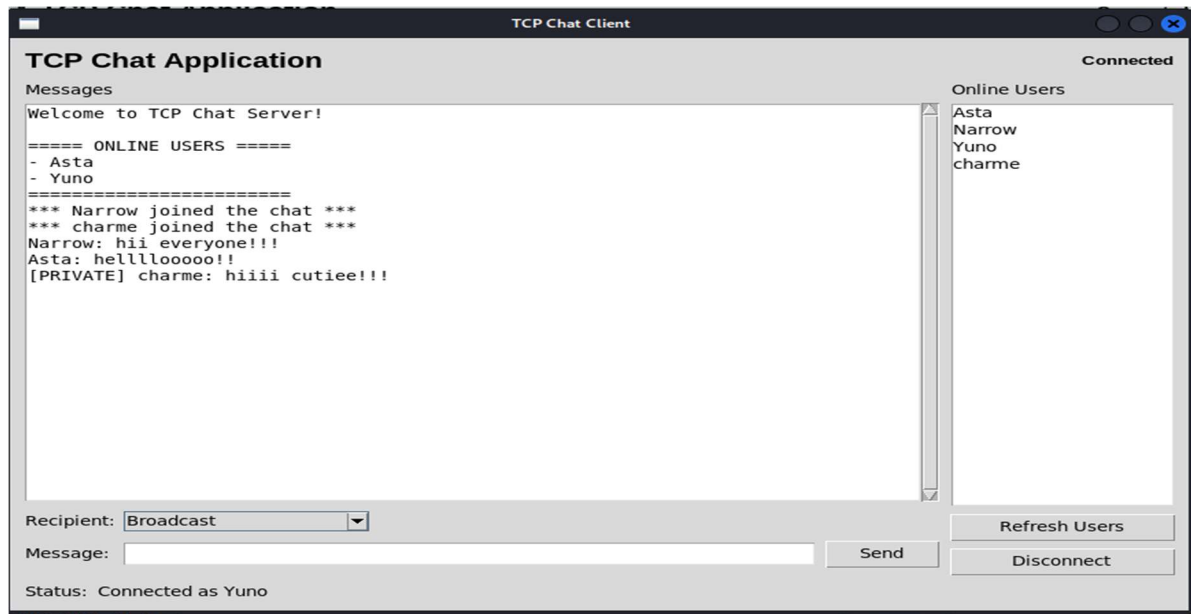


Figure 4: Private messaging.

Disconnect Handling

The application notifies users when clients disconnect and updates the online list.

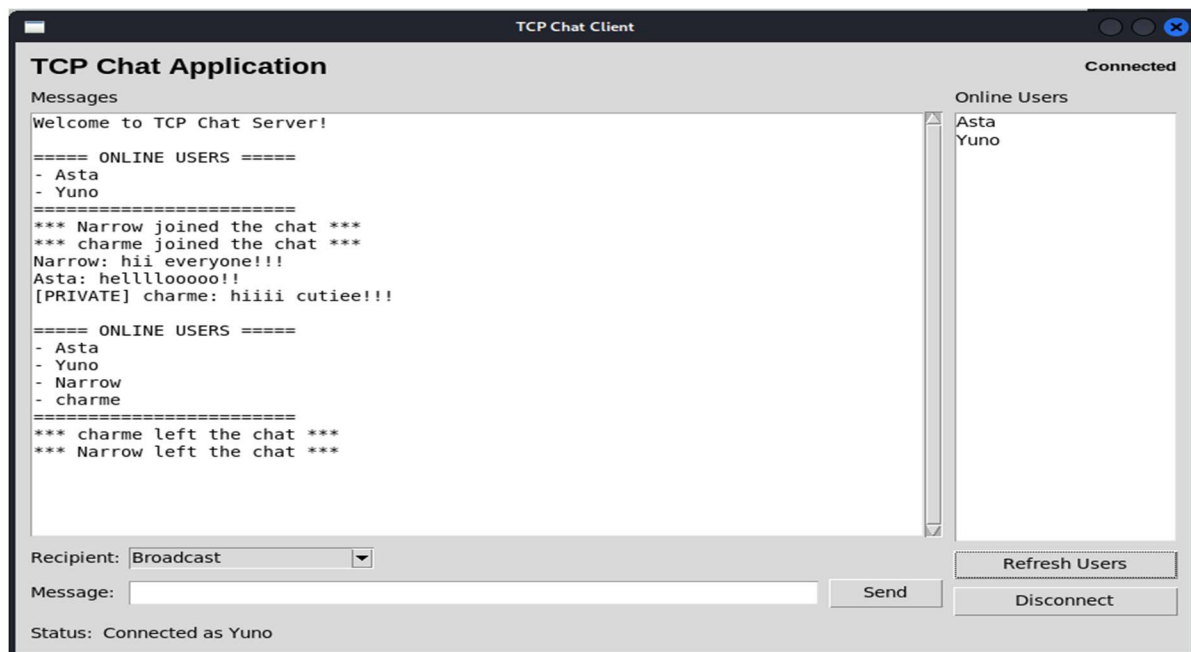


Figure 5: Disconnect handling.

3. System Architecture

The system uses a client-server architecture in Mininet (single,5). Host h1 runs server.py while h2–h5 execute client_gui.py. The GUI communicates with the existing TCP server over port 5000. A background receiver thread updates the interface without blocking user interaction.

4. Components Reused from Assignment 5

- Existing TCP socket communication.
- Threaded server implementation.
- Broadcast messaging.
- Private messaging (/msg).
- Online user listing (/list).
- Chat history logging.
- Join/leave notifications.

5. GUI Design Decisions

Tkinter widgets include Labels, Entry widgets, Buttons, Frames, Listbox, Combobox, ScrolledText and MessageBox. The interface contains a login window followed by a chat window with online users, message area, recipient selection, send/disconnect controls and status indicators.

6. Testing Results

Testing with four Mininet clients confirmed successful login, online user updates, broadcast messaging, private messaging, user join notifications, leave notifications and graceful disconnection. The supplied screenshots demonstrate each feature.

7. Wireshark Verification

TCP packets captured using tcp.port==5000 verify handshake, message delivery and connection termination.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.0.2	10.0.0.1	TCP	74	58986 → 58986 [SYN] Seq=0 Win=42348 Len=0 MSS=1460 SACK_PERM TSval=3833668416 TSecr=0 WS=512
2	0.000055195	10.0.0.1	10.0.0.2	TCP	66	58986 → 58986 [ACK] Seq=1 Ack=1 Win=42496 Len=0 TSval=3833668420 TSecr=2294244517 TSecr=3833668416 WS=512
3	0.002192748	10.0.0.2	10.0.0.1	TCP	82	58986 → 58986 [PSH, ACK] Seq=1 Ack=1 Win=43520 Len=16 TSval=2294244520 TSecr=3833668420
4	0.003147709	10.0.0.1	10.0.0.2	TCP	66	58986 → 58986 [ACK] Seq=1 Ack=17 Win=42496 Len=0 TSval=3833668422 TSecr=2294244520
5	0.004397442	10.0.0.2	10.0.0.1	TCP	70	58986 → 58986 [PSH, ACK] Seq=1 Ack=17 Win=42496 Len=4 TSval=3833668423 TSecr=2294244520
6	0.004481339	10.0.0.2	10.0.0.1	TCP	66	58986 → 58986 [ACK] Seq=17 Ack=5 Win=43520 Len=0 TSval=2294244522 TSecr=3833668423
7	0.004568844	10.0.0.1	10.0.0.2	TCP	95	58986 → 58986 [PSH, ACK] Seq=17 Ack=5 Win=43520 Len=29 TSval=2294244522 TSecr=3833668423
8	0.004595187	10.0.0.1	10.0.0.2	TCP	66	58986 → 58986 [ACK] Seq=5 Ack=46 Win=42496 Len=0 TSval=3833668469 TSecr=2294244522
9	0.050553756	10.0.0.2	10.0.0.1	TCP	186	58986 → 58986 [PSH, ACK] Seq=46 Ack=5 Win=43520 Len=120 TSval=2294244568 TSecr=3833668469
10	0.050587787	10.0.0.1	10.0.0.2	TCP	66	58986 → 58986 [ACK] Seq=5 Ack=166 Win=42496 Len=0 TSval=3833668469 TSecr=2294244568
11	0.050943512	10.0.0.2	10.0.0.1	TCP	71	58986 → 58986 [PSH, ACK] Seq=5 Ack=166 Win=42496 Len=5 TSval=3833668846 TSecr=2294244568
12	0.428281273	10.0.0.2	10.0.0.1	TCP	124	58986 → 58986 [PSH, ACK] Seq=166 Ack=10 Win=43520 Len=58 TSval=2294244946 TSecr=3833668846
13	0.428533138	10.0.0.1	10.0.0.2	TCP	66	58986 → 58986 [ACK] Seq=10 Ack=224 Win=42496 Len=0 TSval=3833668847 TSecr=2294244946
14	0.428626827	10.0.0.2	10.0.0.1	TCP	66	58986 → 58986 [ACK] Seq=10 Ack=224 Win=42496 Len=0 TSval=3833668847 TSecr=2294244946

Figure 6: TCP three-way handshake.

Broadcast packet verification.

```
> Frame 49: Packet, 188 bytes on wire (1504 bits), 188 bytes captured (1504 bits) on interface s1-eth1, 1
> Ethernet II, Src: ca:11:7f:55:78:60 (ca:11:7f:55:78:60), Dst: 92:7c:b7:98:47:31 (92:7c:b7:98:47:31)
> Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.4
> Transmission Control Protocol, Src Port: 5000, Dst Port: 59170, Seq: 46, Ack: 7, Len: 122
> Data (122 bytes)
0000  92 7c b7 98 47 31 ca 11 7f 55 78 60 00 00 45 00  | G1   Ux'  E
0010  00 ae 61 30 40 00 40 06 c5 15 0a 00 00 01 0a 00  | a00@-
0020  00 04 13 88 e7 22 70 4f 47 4c 5f 50 56 22 80 18  | .....x0 GL PV...
0030  00 55 14 a5 00 00 01 01 00 0a 0f ea 7b 99 09 c3  | U.....o [
0040  ab b0 0a 3d 3d 3d 3d 3d 20 59 6f 75 72 20 4c 61  | ..... Your La
0050  73 74 20 35 20 4d 65 73 73 61 67 65 73 20 3d 3d  | st 5 Mes sages ==
0060  3d 3d 3d 0a 32 30 32 36 2d 30 37 2d 30 39 20 30  | == 2026 -07-09 0
0070  37 3a 31 36 3a 32 34 20 7c 20 62 72 6f 61 64 63  | 7:10:24 | broadc
0080  61 73 74 20 7c 20 41 4c 4c 20 7c 20 68 69 69 20  | ast | AL L | hii
0090  65 76 65 72 79 6f 6e 65 21 21 21 0a 3d 3d 3d 3d  | everyone !!! ====
00a0  3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d  | =====
00b0  3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 3d 0a          | ===== ==
```

Figure 7: Broadcast packet.

Private message verification.

```
> Frame 104: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface s1-eth1, id 0
> Ethernet II, Src: f6:e9:56:e7:1c:61 (f6:e9:56:e7:1c:61), Dst: ca:11:7f:55:78:60 (ca:11:7f:55:78:60)
> Internet Protocol Version 4, Src: 10.0.0.5, Dst: 10.0.0.1
> Transmission Control Protocol, Src Port: 46536, Dst Port: 5000, Seq: 28, Ack: 129, Len: 27
> Data (27 bytes)
Data: 2fd736f72059756e6f296809692073770565746965656565212121
[Length: 27]
0000  ca 11 7f 55 78 60 f6 e0 56 e7 1c 61 08 00 45 00  | ..Ux'.. V a E
0010  00 4f cf 84 40 00 40 06 57 0f 0a 00 00 05 0a 00  | . @ @ W .....
0020  00 01 bd 90 13 08 d2 13 cc e9 70 47 d2 ae 00 10  | ..... y .....
0030  00 53 49 7e 00 00 01 01 08 0a f2 be 41 f1 25 0c  | .SI..... A %
0040  2f 23 2f 6d 73 67 20 59 75 6e 6f 20 68 69 69 20  | /#/msg Y uno hii
0050  73 77 65 65 74 69 65 65 65 65 21 21 21          | sweetie ee!!!
```

Figure 8: Private message packet.

TCP FIN/ACK disconnection.

127	198.582460764	10.0.0.5	10.0.0.1	TCP	66	48536 → 5000	[FIN, ACK]	Seq=55 Ack=193 Win=42496 Len=0 TSval=4072570644 TSecr=621584883
128	198.582851877	10.0.0.1	10.0.0.5	TCP	66	5000 → 48536	[FIN, ACK]	Seq=193 Ack=56 Win=43520 Len=0 TSval=621593423 TSecr=4072570644
129	198.582899889	10.0.0.5	10.0.0.1	TCP	66	48536 → 5000	[ACK]	Seq=56 Ack=194 Win=42496 Len=0 TSval=4072570644 TSecr=621593423
130	198.583129993	10.0.0.1	10.0.0.3	TCP	96	5000 → 59368	[PSH, ACK]	Seq=378 Ack=10 Win=43520 Len=30 TSval=3958519439 TSecr=3549358410
131	198.583230420	10.0.0.3	10.0.0.1	TCP	66	59368 → 5000	[ACK]	Seq=10 Ack=408 Win=42496 Len=0 TSval=3549366951 TSecr=3958519439
132	198.583272451	10.0.0.1	10.0.0.4	TCP	96	5000 → 59170	[PSH, ACK]	Seq=326 Ack=12 Win=43520 Len=30 TSval=1877740685 TSecr=183912520
133	198.583354555	10.0.0.4	10.0.0.1	TCP	66	59170 → 5000	[ACK]	Seq=12 Ack=356 Win=42496 Len=0 TSval=163921060 TSecr=1877740685
134	203.195000316	10.0.0.4	10.0.0.1	TCP	66	59170 → 5000	[FIN, ACK]	Seq=12 Ack=356 Win=42496 Len=0 TSval=163925672 TSecr=1877740685
135	203.195265980	10.0.0.1	10.0.0.4	TCP	66	5000 → 59170	[FIN, ACK]	Seq=356 Ack=13 Win=43520 Len=0 TSval=1877745297 TSecr=163925672
136	203.195313485	10.0.0.4	10.0.0.1	TCP	66	59170 → 5000	[ACK]	Seq=13 Ack=357 Win=42496 Len=0 TSval=163925672 TSecr=1877745297
137	203.195557785	10.0.0.1	10.0.0.3	TCP	96	5000 → 59368	[PSH, ACK]	Seq=400 Ack=10 Win=43520 Len=30 TSval=3958524052 TSecr=3549366951
138	203.195661260	10.0.0.3	10.0.0.1	TCP	66	59368 → 5000	[ACK]	Seq=10 Ack=438 Win=42496 Len=0 TSval=3549371564 TSecr=3958524052
139	206.264468045	10.0.0.3	10.0.0.1	TCP	66	59368 → 5000	[FIN, ACK]	Seq=10 Ack=438 Win=42496 Len=0 TSval=3549374633 TSecr=3958524052
140	206.264746915	10.0.0.1	10.0.0.3	TCP	66	5000 → 59368	[FIN, ACK]	Seq=438 Ack=11 Win=43520 Len=0 TSval=3958527121 TSecr=3549374633
141	206.264795020	10.0.0.3	10.0.0.1	TCP	66	59368 → 5000	[ACK]	Seq=11 Ack=439 Win=42496 Len=0 TSval=3549374633 TSecr=3958527121

Figure 9: Disconnect packet.

8. Conclusion

The GUI successfully reuses Assignment 5 networking logic while improving usability through an event-driven interface with background message handling.